

Muhammad Hanif

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EDUCATION

Institute of Science Tokyo (formerly Tokyo Institute of Technology)

Ph.D. in Systems and Control Engineering

Tokyo, Japan

Oct 2022 – Sept 2025

Advisor: Prof. Takeshi Hatanaka

Thesis: Multi-Drone Coordinated Image Sampling for 3D Map Reconstruction through Efficient and Scene-Adaptive Coverage Control

Tokyo Institute of Technology

M.Eng. in Systems and Control Engineering

Tokyo, Japan

Oct 2020 – Sept 2022

Advisor: Prof. Takeshi Hatanaka

Thesis: Real-Time Optimization for Dynamic Multi-Target Allocation & Tracking with Heterogeneous Robot Systems

Bandung Institute of Technology

B.Sc. in Electrical Engineering

Bandung, Indonesia

Aug 2014 – Jul 2018

GPA: 3.75/4.00 | **Advisor:** Prof. Bambang Riyanto & Dr. Egi Hidayat

Thesis: Design and Implementation of Control System in Hybrid Underwater Glider Vehicle in ROS Environment

RESEARCH & INDUSTRY EXPERIENCE

DeepX, Inc.

Robotics Control Engineer

Tokyo, Japan

Nov 2025 – Present

- Designing a coordinated planning algorithm for autonomous excavation using multiple autonomous shovels, contributing to safe and efficient multi-agent operation in soil excavation tasks.

Institute of Science Tokyo / Tokyo Institute of Technology

Research Assistant, Hatanaka Laboratory

Tokyo, Japan

May 2021 – Sept 2025

- Coordinated Multi-Drone Image Sampling and Coverage Control for 3D Reconstruction:** Developed coverage control and image sampling algorithms for UAVs for 3D reconstruction applications. Simulations were conducted in ROS/ROS2 and Unity, with real-world testing in both a controlled testbed and an agricultural farm setting. [J1, J2, J3, C2, C4]
Highlight: ROS/ROS2, Unity, Pytorch, Python, C#, Coverage Control, Control Barrier Function, Multi-UAV, Online Map Feedback, 3D Reconstruction.
- Multi-Robot Task Allocation for Equipment Inspection (in collaboration with Yokogawa Electric):** Developed a simulator using Unity and ROS2 to simulate multi-robot task allocation scenarios in a factory environment. [P1]
Highlight: ROS2, Unity, Python, C#, Path Planning, Multi-Robot Task Allocation.
- Drone Coverage Control for Moving Target Tracking Application (in collaboration with Fujitsu):** Enhanced drone coverage control methods for dynamic target tracking applications by adapting the drone's altitude and detection model in real-time [C3].
Highlight: ROS, Python, Coverage Control, Drone Target Tracking, TensorFlow.
- Safe Autonomous Ship Control Method in a Port (in collaboration with Kawasaki Heavy Industry):** Designed control methods for autonomous ship operation in a port with safety certificates using a combined approach of Model Predictive Control (MPC) and Control Barrier Function (CBF). [B1, C5].
Highlight: MATLAB, Simulink, Control Barrier Function, Model Predictive Control.
- Predictive Multi-Robot Task Allocation for Radiation Monitoring (in collaboration with the University of Seville):** Implemented a high-level control method for multiple heterogeneous robots (UAVs and UGVs) for solar irradiance monitoring tasks in a thermosolar plant. [J4, C7, D5]
Highlight: ROS, Gazebo, Python, Optimization-based Control.

FBTriangle

Simulation Software Engineer Intern

Tokyo, Japan

Jan 2025 – Mar 2025

- Developed a coverage control algorithm to simulate optimal defensive formations in ice hockey, contributing to realistic multi-agent behavior in sports strategy software.

IROS Tech

Drone Engineer

Bandung, Indonesia

Apr 2019 – May 2020

- Developed UAV systems for autonomous path planning to support mapping and surveillance tasks, including generating 3D photogrammetry maps for inspection purposes.

Highlight: 3D Reconstruction, Photogrammetry, Mapping, UAV.

Bandung Institute of Technology

Research Assistant, Advanced Robotics Research Laboratory

Bandung, Indonesia

July 2018 – Apr 2019

- **Design & Development of Autonomous Hybrid Underwater Glider:** Designed and developed autonomous Hybrid Underwater Vehicle (HUG) for long-distance surveillance missions and conducted lake testing. Led the design team to iterate all engineering aspects of the vehicle, including software, hardware, and mechanical components [T3,P2].

Highlight: Hybrid Underwater Glider (HUG), Waypoint Tracking, ROS, C++, Python.

Research Assistant, Biomedical Research Group

May 2018 – Nov 2018

- **Automatic Diabetic Retinopathy Classifier:** Built algorithms and machine learning models using TensorFlow for automatic grading of diabetic retinopathy severity [C8].
- **CNN-based Brain Tumor Classifier:** Developed automatic brain tumor classification from a brain tumor dataset into three classes using deep learning [C9].

Highlight: Machine Learning, Xception, CNN, Tensorflow, Keras, OpenCV.

Research Assistant, CentrUMS (Center for Unmanned System Studies)

Aug 2016 – Nov 2018

- **Development of High Altitude Long Endurance (HALE) UAV:** Implemented a control and monitoring system for a HALE (High Altitude Long Endurance) UAV with a wingspan of 21 meters. Conducted several flight tests, including testing a mini prototype UAV at an altitude of 10,000 feet.

Highlight: Pixhawk, Mission Planner, Waypoint Tracking, Real-Time Monitoring Streaming, Antenna Tracker.

Researcher, Aksantara Drone Research Team

Aug 2015 – June 2017

- **Design & Development of Tube-launched Folding-Wing UAV:** Built control and guidance systems for the folding-wing tube-launched drone. Achieved best design recognition at the national drone competition in 2017 [C10].
- **Hybrid Vertical Take-Off Landing (VTOL) UAV :** Built multifunctional Hybrid VTOL (Vertical Take-Off and Landing) UAV. Led a team of 10 diverse engineers to develop the UAV system from scratch, achieving 1st Runner-Up in the National Drone Competition in 2016.

Highlight: Pixhawk, Mission Planner, Hybrid VTOL, Multirotor, Fixed-Wing

Aero Terra Scan

Software Developer Intern

Bandung, Indonesia

June 2017 – Aug 2017, Internship

- Designed and implemented the back-end program for the Ground Control System (GCS) for UAV (Unmanned Aerial Vehicle) mapping missions using C#.

Highlight: Mission Planner, Visual Studio, C#, Pixhawk

Book, Book Chapter, Editing

- [B1] Otsuki, Satoshi, Naoki Hatta, Muhammad Hanif, Riku Funada, Kenichi Nakashima, and Takeshi Hatanaka. "**Hierarchical Vessel Safe Operation in A Port through CBF, MPC and RRT-like Spatiotemporal Path Planning**" In *Nonlinear and Constrained Control - Applications, Synergies, Challenges and Opportunities*, E. Garone, I.V. Kolmanovsky, and T.W. Nguyen (eds), Springer Nature, pp. 407-441 (2025)

Journal Papers

- [J1] Muhammad Hanif, T. Sumino, K. Uto, D. Ichihashi, K. Cheng, and T. Hatanaka. "**Coverage-Recon: Coordinated Multi-Drone Image Sampling with Online Map Feedback.**" *IEEE Transactions on Control Systems Technology*, under review (2025).
- [J2] Muhammad Hanif, Takumi Shimizu, Zhiyuan Lu, Masaya Suenaga, and Takeshi Hatanaka. "**Efficient Angle-Aware Coverage Control for Large-Scale Map Reconstruction using Drone Networks.**" *SICE Journal of Control, Measurement, and System Integration* 17, no. 1 (2024): 144-155 [[Paper](#)] [[Video](#)]
- [J3] Lu, Zhiyuan, Muhammad Hanif, Takumi Shimizu, and Takeshi Hatanaka. "**Angle-Aware Coverage with Camera Rotational Motion Control.**" *SICE Journal of Control, Measurement, and System Integration* 17, No. 1 (2024): 211–221 [[Paper](#)]
- [J4] Martin, J. G., Muhammad Hanif, T. Hatanaka, J. M. Maestre, and E. F. Camacho. "**Predictive receding-horizon multi-robot task allocation applied to the mapping of direct normal irradiance in a thermosolar power plant.**" *Solar Energy* 263 (2023): 111911. [[Paper](#)] [[Video](#)]

International Conference Papers

- [C1] K. Uto, Muhammad Hanif, S. Takahashi, M. Takahashi, K. Yoshi, and T. Hatanaka. "**Hyperspectral Data-Based Growth State Estimation of Rice Plants Cultivated in Closed Plant Factories with Artificial Lighting.**" *European Plant Phenomics Symposium (EPPS)*, 2025.
- [C2] K. Uto, S. Takahashi, Muhammad Hanif, M. Takahashi, K. Yoshi, and T. Hatanaka. "**Hyperspectral Imaging for Useful Substance Production Using Rice Plants Cultivated in Closed Plant Factories with Artificial Lighting.**" *3rd BrIAS Conference on Smart Agriculture*, Brussels, Belgium, February 4–6, 2026.
- [C3] M. Suenaga, Muhammad Hanif, K. Uto, and T. Hatanaka. "**Hierarchical Multi-Robot Data Sampling for Environmental State Estimation through Online Gaussian Process.**" *Proc. 2025 European Control Conference*, (2025)
- [C4] Muhammad Hanif, T. Sumino, K. Uto, D. Ichihashi, K. Cheng, and T. Hatanaka. "**Impact of Real-time Map Feedback on Coordinated Image Sampling for 3D Reconstruction.**" *Proc. 2025 European Control Conference*, (2025) [[Paper](#)] [[Video](#)]
- [C5] Muhammad Hanif and Takeshi Hatanaka. "**Real-time Adaptation of Drone Altitude and Object Detection Model for Moving Target Tracking.**" *Proc. SICE Annual Conference*, pp. 533-538 (2024) [[Paper](#)]
- [C6] Lu, Zhiyuan, Muhammad Hanif, and Takeshi Hatanaka. "**Angle-Aware Full 3D Coverage Control with ADMM-based Dynamic Assignment of Charging Stations.**" *Proc. SICE Annual Conference*, pp. 1119-1124 (2024) [[Paper](#)]
- [C7] Otsuki, Satoshi, Naoki Hatta, Muhammad Hanif, Takeshi Hatanaka, and Kenichi Nakashima. "**Hierarchical Vessel Autonomous Operation in a Port with Safety Certificates: Combined MPC and CBF Approach.**" In *IFAC-PapersOnLine* 56, no. 2 (2023): 3138-3145. [[Paper](#)] [[Video](#)]

- [C8] Asavasirikulkij, Chanun, and Muhammad Hanif. "**Human Workload Evaluation of Drone Swarm Formation Control using Virtual Reality Interface.**" In *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*, pp. 132-136. 2023. [\[Paper\]](#) [\[Video\]](#)
- [C9] Martin, Javier G., Muhammad Hanif, Takeshi Hatanaka, Jose M. Maestre, and Eduardo F. Camacho. "**Predictive receding-horizon multi-robot task allocation with moving tasks.**" In *2022 European Control Conference (ECC)*, pp. 2030-2035. IEEE, 2022. [\[Paper\]](#)
- [C10] Lazuardi, Rachmadio Noval, Nyoman Abiwinanda, Tafwida Hesaputra Suryawan, Muhammad Hanif, and Astri Handayani. "**Automatic diabetic retinopathy classification with efficientnet.**" In *2020 IEEE REGION 10 CONFERENCE (TENCON)*, pp. 756-760. IEEE, 2020. [\[Paper\]](#)
- [C11] Abiwinanda, Nyoman, Muhammad Hanif, S. Tafwida Hesaputra, Astri Handayani, and Tati Rajab Mengko. "**Brain tumor classification using convolutional neural network.**" In *World Congress on Medical Physics and Biomedical Engineering 2018: June 3-8, 2018, Prague, Czech Republic (Vol. 1)*, pp. 183-189. Springer Singapore, 2019. [\[Paper\]](#)
- [C12] Ahmad Fadlillah Muzammil, Nurhayyan Halim Rosid, Muhammad Hanif, Naufalino Fadel, Nathan, Tobias S., Tegar S., M.Agoes Moelyadi, and Agus Budiyo. "**Design and Development of Tube-Launched Unmanned Aerial Vehicle.**" In *International Conference on Intelligent Unmanned Systems*, August, 2018. [\[Paper\]](#)

Domestic Conferences & Invited Talks

- [D1] Muhammad Hanif. "**Orchestrating Multi-Robot Collaboration: From Aerial Drones to Heavy Machinery.**" Plenary Speaker, *The 17th Multidisciplinary International Student Workshop (MISW 2026)*, Institute of Science Tokyo, Tokyo, Japan, August 3-4, 2026.
- [D2] Muhammad Hanif. "**Drones Meet 3D-AI: Multi-Drone Collaboration for 3D Reconstruction.**" Tokyo AI x Drones – Invited Session, Tokyo, May 30, 2025. [\[Slides\]](#)
- [D3] Takumi Sumino, Muhammad Hanif, Kuniaki Uto, Daisuke Ichihashi, K. Cheng, Takeshi Hatanaka. "**Feedback-Based Coverage Control Using Reconstructed Models from Real-Time 3D Mapping.**" *The 12th SICE Control Division Multi-Symposium, Osaka, 2025*.
- [D4] Muhammad Hanif. "**Empowering Precision Agriculture: Efficient Angle-Aware Coverage for 3D Structure Reconstruction with Drone Teams.**" SSS Global Forum 2023: Future of Super Smart Society, Tokyo Institute of Technology, November 3, 2023. [\[Poster\]](#)
- [D5] Muhammad Hanif. "**Human-collaborative Control of Drone Networks with Virtual Reality.**" SSS Matching Workshop, Tokyo Institute of Technology, November 16, 2022. [\[Poster\]](#)
- [D6] Muhammad Hanif. "**Computationally Inexpensive UAV/UGV Allocation for Radiation Monitoring over Mega Solar Power Plants.**" SSS Matching Workshop, online, June 9, 2021.

Thesis

- [T1] Muhammad Hanif. "**Multi-Drone Coordinated Image Sampling for 3D Map Reconstruction through Efficient and Scene-Adaptive Coverage Control.**" Ph.D. Dissertation, Department of Systems and Control Engineering, Institute of Science Tokyo, July 2025.
- [T2] Muhammad Hanif. "**Real-Time Optimization for Dynamic Multiple Target Allocation and Tracking with Heterogeneous Robotic Systems.**" M.Eng. Thesis, Department of Systems and Control Engineering, Tokyo Institute of Technology, Aug 2022. [\[Thesis\]](#)
- [T3] Muhammad Hanif. "**Design and Implementation of Control System in Hybrid Underwater Glider Vehicle in ROS Environment.**" B.Sc. Thesis, Department of Electrical Engineering, Bandung Institute of Technology, July 2018. [\[Thesis\]](#)

Patents

- [P1] Hiroki Sugawara, Yasuki Sakurai, Muhammad Hanif, Yuya Okada, Takeshi Hatanaka, “**Robot Operation System**,” US Patent Application (Under Review), October 24, 2024.

Description: Proposed a robot operation and task allocation system designed for long-term operations while optimizing task assignments, collision avoidance, and motion planning. The system integrates facility management data, external inputs, and scheduling to ensure efficient operations within the specified time. It also pre-computes motion paths to account for environmental constraints, enabling safe and effective deployment of robots.

- [P2] Sapto Adi Nugroho, Natsir Habibullah, Simon Siregar, Bambang Riyanto Trilaksono, Egi Muhammad Idris Hidayat, Fachry Ahmad Efendi Yakin, Muhammad Hanif, Muhammad Faisal Sagala, et al., “**Buoyancy Engine using Fluid Bags on a Hybrid Autonomous Underwater Glider and Method for Operating It**,” Indonesia Patent No. P00202102652, April 13, 2021. [\[Patent\]](#)

Description: Invented a buoyancy engine for a Hybrid Autonomous Underwater Glider (HAUG) using fluid bags and described methods for its operation. The system uses a hydraulic pump, valves, sensors, and internal and external fluid bags to control buoyancy.

AWARDS & RECOGNITION

WISE SSS Doctoral Leadership Scholarship: Funding support awarded by WISE SSS Tokyo Tech for enrolled students with high research ability and outstanding potential (Oct 2023 – Sept 2025)

Japanese Government MEXT Scholarship: Scholarship awarded by Japanese Government to conduct graduate study in Japan (Oct 2020 – Sept 2025)

Indonesia Aerial Robotics Competition 2017 - Best Design: Won Best Design awards in the national aerial robotics competition for building Folding-Wing Tube-Launched UAV (Sept 2017)

Smart City Robot Innovation Challenge - Champion: Won the first champion in the domestic robotics competition for developing Vertical Take-Off Unmanned Aerial Vehicle (Apr 2017)

Indonesia Aerial Robotics Competition 2016 - 1st Runner Up: Won 1st Runner-Up Champion in the national aerial robotics competition for building Hybrid VTOL UAV (Sept 2016)

Bandung Institute of Technology - Outstanding Student Award: Awarded to students who have been to be the top 1% for the academic performance among the year. (Aug 2015)

Indonesia National Science Olympiad 2013 - Bronze Medal Won bronze medal at the Indonesia National Science Olympiad 2013 in the field of mathematics. (Aug 2013)

International Young Mathematician Convention India 2012 - Silver Medal Won silver medal at the International Mathematics Competition in India 2012 (Nov 2012)

International Mathematics Contest Singapore 2011 - Silver Medal Won silver medal at the International Mathematics Contest Singapore in 2011 (Aug 2011)

Invitational World Youth Mathematics Intercity Competition 2011 - Bronze Medal Won bronze medal at the IWYMIC Indonesia in 2011 at the Individual Contest (July 2011)

Invitational World Youth Mathematics Intercity Competition 2010 - Bronze Medal Won bronze medal at the IWYMIC South Korea in 2010 at the Team Contest (July 2010)

TEACHING EXPERIENCE

Department of System and Control Engineering, Tokyo Institute of Technology

- **Optimal Control** by Prof. Takeshi Hatanaka, Teaching Assistant (Spring 2024)
- **Cyber Physical Innovation**, Teaching Assistant (Fall 2022)

Department of Electrical Engineering, Bandung Institute of Technology

- **Image Processing** by Prof. Tati Rajab Mengko & Dr. Astri Handayani, Teaching Assistant (Fall 2018)

Mathematics Education Clinic, Bogor, Indonesia

- **Math Olympiads for Elementary and Middle Schools**, Tutor (2012 - 2014)

TECHNICAL SKILLS

Languages: English (Proficient C1), Japanese (Intermediate N3), Arabic (Intermediate), Indonesian (Native)

Programming: C/C++, C#, Python, MATLAB, JavaScript, L^AT_EX

Software: Systems (Linux, Windows, ROS/ROS2), Pytorch, Tensorflow, CUDA, OpenCV, Git, Visual Studio, Unity, Gazebo, Eagle, Altium Designer, Mission Planner, Ardupilot

Hardware: Raspberry-Pi, Arduino, Beaglebone, Udoo, Pixhawk, Ardupilot, DJI Mavic, Parrot Bebop

SELECTED RESEARCH HIGHLIGHTED IN MEDIA

- Article: "[ARNADYAKSA, Mini ‘Submarine’ by ITB Students](#)", by Ahmad Fadil, ITB News, 2018
- Article: "[Mahasiswa ITB Ciptakan Kapal Selam Mini Tanpa Awak](#)", by Kumparan News, 2018
- Article: "[Mahasiswa ITB Buat Robot Pesawat yang Sayapnya Bisa Dilipat](#)", by Kumparan News, 2017
- News: "[Nirwak Pesawat Lipat Karya Mahasiswa ITB](#)", by detikcom, 2017

EXTERNAL LINKS

- Website: <https://mhd-hanif.github.io>
- Google Scholar: <https://bit.ly/hanif-scholar>
- Github: <https://github.com/mhd-hanif>
- LinkedIn: <https://www.linkedin.com/in/muhammad-hanif-6189b0157/>